

Homework 10: Nonnegative Matrix Factorization

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Problem 24: Uniqueness of NMF solutions

The factor matrices W and H in a non-negative matrix factorization $V \approx WH$ are **not unique**. For any invertible $K \times K$ matrix Q such that both Q and Q^{-1} have non-negative entries, the matrices

$$\tilde{W} = WQ \in \mathbb{R}_{\geq 0}^{M \times K}, \quad \tilde{H} = Q^{-1}H \in \mathbb{R}_{\geq 0}^{K \times N}$$

satisfy $\tilde{W}\tilde{H} = WQQ^{-1}H = WH$ but in general $\tilde{W} \neq W$ and $\tilde{H} \neq H$. Such matrices Q are the monomial matrices, i.e. products of a permutation matrix and a positive diagonal matrix.

Concrete counter-example, diagonal scaling Let $D = \text{diag}(d_1, \dots, d_K)$ with $d_k > 0$ and not all $d_k = 1$. Then

$$\tilde{W} := WD, \quad \tilde{H} := D^{-1}H$$

are non-negative because W, H are non-negative and D, D^{-1} are positive diagonal, differ from W and H , and

$$\tilde{W}\tilde{H} = (WD)(D^{-1}H) = W(DD^{-1})H = WH.$$

For example with $K = 2$, taking

$$W = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix},$$

gives $\tilde{W} = \begin{pmatrix} 2 & 1 \\ 2 & 2 \end{pmatrix}$ and $\tilde{H} = \begin{pmatrix} 1/2 & 1 \\ 1 & 1 \end{pmatrix}$, which are still non-negative and yield the same product $WH = \tilde{W}\tilde{H} = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$.

Permutation ambiguity Likewise, for any $K \times K$ permutation matrix P which satisfies $P^{-1} = P^T \geq 0$, the pair $\tilde{W} = WP$, $\tilde{H} = P^TH$ reorders the components without changing the product. Combining the two yields the full monomial ambiguity $\tilde{W} = WPD$, $\tilde{H} = D^{-1}P^TH$.

Hence NMF factors are unique only up to a permutation and rescaling of the components. Beyond these trivial ambiguities, additional non-uniqueness can occur for specific data, but the simple counter-examples above already suffice to disprove uniqueness in general.